Dynamical Systems

Assignment 1

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In this report, I will begin to analyze a toy model of glycemic control.

We begin with:

$$u = \alpha v + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') dt', \text{ and}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t),$$

where

- v is the deviation from the baseline glucose concentration.
- Glucose is used at a rate γ by basic functions.
- F(t) can be positive (food intake) or negative (exertion).
- The control variable u serves to keep v at a reasonable level.

1 Step 1: Deriving an ODE with 2 DoF and nondimensionalizing:

Using the Leibniz rule,

$$\dot{u} = \alpha \dot{v} + \beta\lambda v - \beta\lambda \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') dt'.$$

Then

$$\dot{u} + \lambda u = \alpha \dot{v} + \beta\lambda v + \alpha\lambda v$$

$$\dot{u} = \lambda(\beta + \alpha)v - \lambda u + \alpha f(u, v), \text{ where } f(u, v) = \dot{v} \text{ and,}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t)$$

Now let $v = \bar{v}v'$, $u = \tilde{u}u'$, and $t = \tilde{t}t'$, $\dot{v}$ becomes

$$\frac{1}{\tilde{t}} \bar{v} \frac{dv'}{dt'} = -\gamma(\bar{v}v') - \delta(\tilde{u}u')(\bar{v}v' + \bar{v}) + F$$

$$\frac{dv'}{dt'} = -(\gamma\tilde{t})v' - (\delta\tilde{t}\tilde{u})u'(v' + 1) + \frac{\tilde{t}}{\bar{v}} F.$$

$\tilde{t}$ and $\tilde{u}$ can be chosen such that

$$\gamma\tilde{t} = 1 \longrightarrow \tilde{t} = \frac{1}{\gamma}, \text{ and}$$

$$\delta\tilde{t}\tilde{u} = 1 \longrightarrow \tilde{u} = \frac{1}{\delta\tilde{t}} = \frac{\gamma}{\delta}. \text{ Let } \frac{\tilde{t}}{\bar{v}} F = G \text{ and we have}$$

$$\frac{dv'}{dt'} = -v' - u'(v' + 1) + G = f'(u', v'). \text{ This is now dimensionless.}$$

Making the same substitutions in $\dot{u}$, we get

$$\frac{du'}{dt'} = -au' + bv' + cf'(u', v'), \text{ where } a = \frac{\lambda}{\gamma}, b = \frac{\lambda\bar{v}}{\delta}(\alpha + \beta), c = \frac{\alpha}{\delta}, \text{and, } f'(u', v') = \frac{dv'}{dt'}.$$

2 Finding the equilibria:

Set $\dot{v} = \dot{u} = 0$.

$$-v - u(v+1) + G = 0$$

$-au + bv = 0 \longrightarrow u = \frac{b}{a}v$. Let $k = b/a$. Then $u = kv$. Plug into equation above to get

$$-v - kv(v+1) + G = 0, \text{ rearranging}$$

$$kv^2 + (1+k)v - G = 0.$$

$$v = \frac{-(1+k) \pm \sqrt{(1+k)^2 + 4kG}}{2k}, u = kv.$$

For real roots as G is changed

$G = 0$	$G > G^*$	$G < G^*$
$v = 0, v = \frac{-(1+k)}{k}$	$v = \frac{-(1+k) \pm \sqrt{(1+k)^2 + 4kG}}{2k}$	N/A

with $G^* = \frac{-(1+k)^2}{4k}$, found by setting the square root to 0.

For $G > G^*$ these solutions look like

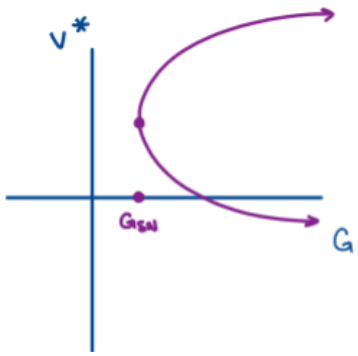


Figure 1: Equilibrium solutions vs G

The position and eccentricity of the graph may change as k changes but there is always a saddle node at G^* and real solutions only exist for $G > G^*$.

3 Experimenting with parameters

3.1 Changing parameters

Now I will present results from experiments I did with the parameter G to see how the controllability of the model changes as the parameters are changed.

For this experiment I used the file exp1-equilibria.py. I changed the parameters a, c, and b to see if I could find a combination where the equilibrium branch is unstable. The table below summarizes the results.

$(a,\ c,\ b)$	$k = b/a$	G^*	Plus branch unstable for G in
(1, 1, 2)	2	−1.125	never
(1, 1, 1)	1	−1.000	never
(1, 10, 5)	5	−1.800	never
(1, 1, 5)	5	−1.800	$[-1.800, -1.750]$
(0.1, 1, 1)	10	−3.025	$[-3.025, -1.736]$

Table 1: Controllability for a constant input G switched on at the baseline state $(u, v) = (0, 0)$. For $G < G^*$ no equilibrium exists and the orbit crashes in every case. G was scanned from −4 to 3 in steps of 0.05. The values of G remained consistent as the run time was increased as well as when the step size was reduced.

The figure below shows the equilibrium curve for the last row in Table 1.

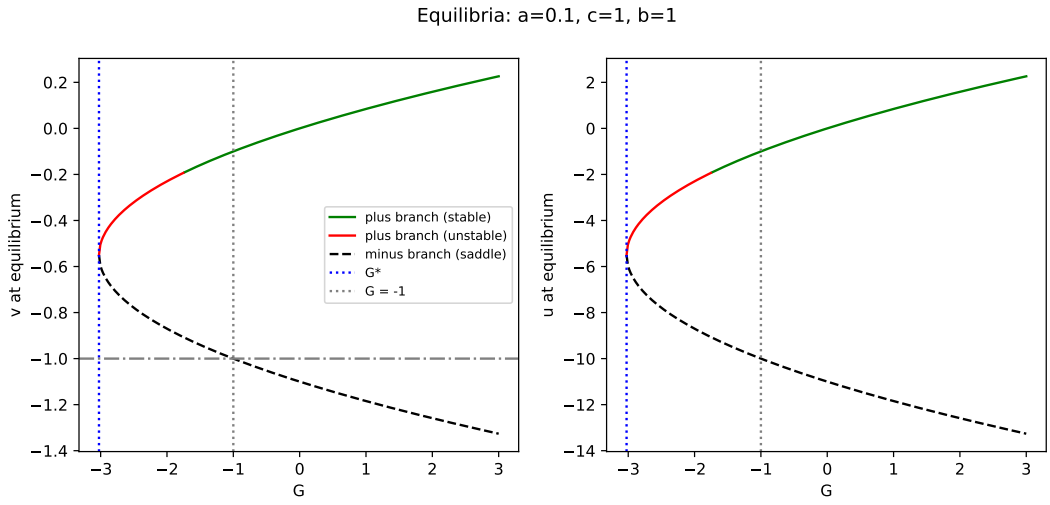


Figure 2: Equilibrium curve for (a,c,b) = (0.1, 1, 1). The red and green lines represent the plus branch, i.e. both eigenvalues have the same sign. The difference is in the trace of the Jacobian. If the trace is negative, the branch is stable in that area, and if the trace is positive, the branch is unstable. At the boundary of red/green, the trace is 0.

3.2 Changing ICs

Initial (u_0, v_0)	Description	Lowest G that converges	Outcome above threshold
$(-5,\ 0)$	control pre-suppressed	-1.5	controlled
$(0,\ 1)$	high glucose (after a meal)	-1.375	controlled
$(0,\ 0)$	baseline	-1.234	controlled
$(0,\ -0.5)$	low glucose	-1.127	controlled
$(5,\ 0)$	strong control already active	-0.9933	controlled
$(0,\ -1.5)$	$v_0 < -1$ (negative glucose)	never	crashed
$(-8,\ -1.8)$	$v_0 < -1$, control strongly negative	never	crashed

Table 2: Dependence of controllability on the initial condition for fixed parameters $(a,\ c,\ b) = (0.1,\ 1,\ 1)$, for which the saddle node point is at $G^* = -3.025$ and the plus branch is unstable for $G \in [-3.025,\ -1.736]$.

Table 2 above shows some realistic examples of what happens with different initial conditions of the glucose control parameter (u) as well as the glucose level itself (v).

From the lecture, we know that negative u means the control variable seeks to increase glucose levels, and the opposite is true when u is positive. We also know that negative G means that variable is working to reduce glucose levels. The more negative G is the more exhaustion of glucose.

So, analyzing the data in the table, the results seem pretty realistic. When the initial glucose levels are at a baseline ($v = 0$), the control variable u determines how strong G is. When the initial glucose levels are high, with the u being low, G has to be strongly negative to get back to baseline. The initial glucose levels can't really be negative in reality but I tried some runs anyway. They all failed to converge at any G. I tried expanding the range of G, as well as changing the step size, and several different values for both u_0 and v_0.